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System and method for sending, delivery and receiving of faxes using a remotely deployed fax service or system

Abstract

Embodiments of systems and methods for the sending, delivery and receiving of faxes are disclosed herein. In particular, certain embodiments include a fax connector that may be deployed at a user's site, and a faxing system that may be deployed remotely from the user's site. The fax connector can be accessed at the user's site over a computer based network in order to perform functions associated with faxing, including sending, receiving and obtaining status on faxes. The fax connector deployed at the user's site communicates with the remotely deployed faxing system to send, receive, obtain status on, or perform other functions associated with, faxing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This patent application is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 18/080,234 filed Dec. 13, 2022, issued as U.S. Pat. No. 11,843,750, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES USING A REMOTELY DEPLOYED FAX SERVICE OR SYSTEM,” which is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 17/826,562, filed May 27, 2022, issued as U.S. Pat. No. 11,558,532, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES USING A REMOTELY DEPLOYED FAX SERVICE OR SYSTEM,” which is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 17/131,516, filed Dec. 22, 2020, issued as U.S. Pat. No. 11,375,083, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES USING A REMOTELY DEPLOYED FAX SERVICE OR SYSTEM,” which is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 16/900,456, filed Jun. 12, 2020, issued as U.S. Pat. No. 10,911,637, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES USING A REMOTELY DEPLOYED FAX SERVICE OR SYSTEM,” which is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 16/555,906, filed Aug. 29, 2019, issued as U.S. Pat. No. 10,728,421, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES USING A REMOTELY DEPLOYED FAX SERVICE OR SYSTEM,” which is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 16/025,168, filed Jul. 2, 2018, issued as U.S. Pat. No. 10,440,228, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES USING A REMOTELY DEPLOYED FAX SERVICE OR SYSTEM,” which is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 15/356,078, filed Nov. 18, 2016, issued as U.S. Pat. No. 10,097,728, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES,” which is a continuation of, and claims a benefit of priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 14/556,958, filed Dec. 1, 2014, issued as U.S. Pat. No. 9,544,469, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES,” which is a continuation of U.S. patent application Ser. No. 13/461,351, filed May 1, 2012, issued as U.S. Pat. No. 8,937,736, entitled “SYSTEM AND METHOD FOR SENDING, DELIVERY AND RECEIVING OF FAXES,” which are hereby fully incorporated by reference herein for all purposes.

TECHNICAL FIELD

(1) This disclosure relates generally to digital content delivery in a network environment. More particularly, this disclosure relates to the transmission, delivery or reception of faxes through network based protocols or telephone networks. Even more specifically, this disclosure relates to the transmission, delivery or reception of faxes using distributed services.

BACKGROUND

(2) A facsimile (fax) comprises a document in a digital image format. The document is associated with metadata that specifies a destination with respect to the publicly switched telephone network (PSTN). Thus, typically for at least a portion of the transmission, the document is transmitted over the PSTN. Faxing provides a secure, traceable and non-repudiable mechanism for content delivery and can be particularly useful in business environments.

(3) Faxing, however, has limitations. For example, conventional fax machines can only transmit black and white files with limited resolution in a specific format such as the Tagged Image File

Format (TIFF). Further, the transmission speed can be very slow compared to today's Internet based standards. Not surprisingly, the faxing technology in recent years has faced increasing competition from Internet Protocol (IP) network (e.g. Internet, LANs, WANs, intranets, extranets, etc.) based alternatives (generally referred to as Internet faxing). For example, instead of standalone fax machines, fax servers may be utilized in enterprise network environments. Typically, a fax server is a server computer having the necessary fax software and hardware to receive and send faxes over IP networks. Using fax servers may reduce the cost of transmission since there is no telephone connection charge when sending a fax over the Internet.

(4) Internet based alternatives to physical fax machines are not without drawbacks. One challenge is that the Internet carries a vast range of information resources and services over a hodge-podge of protocols. For example, a user may send an email over the Internet, upload a file to a network site, and make a phone call over the Internet. The email may be sent via the Simple Mail Transfer Protocol (SMTP), the file may be uploaded via the File Transfer Protocol (FTP), and the phone call may be conducted using the Voice over IP (VoIP). Many existing network protocols do not meet the secure, traceable and non-repudiable needs of today's businesses. For example, the email (along with any attachment) as well as the file sent over the Internet unencrypted can be vulnerable to interception.

(5) Additionally, network faxing solutions may include the need to access fax modems, routers or the PSTN (e.g. telephone lines) at the user's physical location. Increasingly, however, in most other areas of communication, including telephony itself, the trend has been toward use of the IP network based protocols (e.g. SMTP, FTP, VoIP, etc.). Accordingly, many traditional fax users no longer wish to maintain telephone lines or other access to the PSTN. It would therefore be desirable to give these users the ability to continue sending and receiving faxes without having to maintain local access to the PSTN.

SUMMARY

(6) To address these needs, among others, embodiments of systems and methods for the sending, delivery and receiving of faxes are disclosed herein. In particular, certain embodiments include a fax connector that may be deployed at a user's site, and a faxing system that may be deployed remotely from the user's site. The fax connector can be accessed at the user's site over a computer based network in order to perform functions associated with faxing, including sending, receiving and obtaining status on faxes. The fax connector deployed at the user's site communicates with the remotely deployed faxing system to send, receive, obtain status on, or perform other functions associated with, faxing.

(7) Because, in most cases, fax users already perform faxing utilizing fax servers, fax appliances, fax software, etc. connected to an computer based network (e.g. an IP (or other protocol) based network), faxing can be accomplished using embodiments of such systems and methods substantially without deviation from the user's previous faxing methodologies or workflows. Thus, such users can use their familiar fax environment without the restrictions of using a browser and without the need to maintain local telephony hardware, licenses, or telephone lines.

(8) In certain embodiments, a set of services deployed on a cloud computing platform may be configured to receive a request over a computer based network to send a fax from a fax connector at a user's site, receive the document and destination telephone number associated with the fax and send the fax over the publicly switched telephone network (PSTN) using a fax service.

(9) In one embodiment, the fax service may send a request to the set of services to determine if there are any faxes to send. The set of services may receive the request, determine that the fax is to be sent and return the document to the fax service over the computer based network.

(10) One or more embodiments may receive a request for a status of the fax from the fax connector and return the status to the fax connector over the computer based network. The status may be updated by the fax service over the computer based network.

(11) In other embodiments, the fax service may receive a fax over the PSTN, and send the

document associated with the received fax to the set of services which determines a fax connector identifier to associate with the received document and sends the document to the fax connector at the user's site associated with the fax connector identifier over the computer based network.

(12) These, and other, aspects of the invention will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. The following description, while indicating various embodiments of the invention and numerous specific details thereof, is given by way of illustration and not of limitation. Many substitutions, modifications, additions or rearrangements may be made within the scope of the invention, and the invention includes all such substitutions, modifications, additions or rearrangements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings accompanying and forming part of this specification are included to depict certain aspects of the invention. A clearer impression of the invention, and of the components and operation of systems provided with the invention, will become more readily apparent by referring to the exemplary, and therefore nonlimiting, embodiments illustrated in the drawings, wherein identical reference numerals designate the same components. Note that the features illustrated in the drawings are not necessarily drawn to scale.

(2) FIG. 1 depicts one embodiment of an architecture for a faxing system.

(3) FIG. 2 depicts one embodiment of an architecture for a faxing system.

(4) FIGS. 3A1-3B depict one embodiment for sending or receiving a fax.

DETAILED DESCRIPTION

(5) The invention and the various features and advantageous details thereof are explained more fully with reference to the nonlimiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known starting materials, processing techniques, components and equipment are omitted so as not to unnecessarily obscure the invention in detail. It should be understood, however, that the detailed description and the specific examples, while indicating preferred embodiments of the invention, are given by way of illustration only and not by way of limitation. Various substitutions, modifications, additions and/or rearrangements within the spirit and/or scope of the underlying inventive concept will become apparent to those skilled in the art from this disclosure. Embodiments discussed herein can be implemented in suitable computer-executable instructions that may reside on a computer readable medium (e.g., a hard disk (HD)), hardware circuitry or the like, or any combination.

(6) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, article, or apparatus that comprises a list of elements is not necessarily limited only those elements but may include other elements not expressly listed or inherent to such process, product, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

(7) Additionally, any examples or illustrations given herein are not to be regarded in any way as restrictions on, limits to, or express definitions of, any term or terms with which they are utilized. Instead, these examples or illustrations are to be regarded as being described with respect to one particular embodiment and as illustrative only. Those of ordinary skill in the art will appreciate that any term or terms with which these examples or illustrations are utilized will encompass other embodiments which may or may not be given therewith or elsewhere in the specification and all such embodiments are intended to be included within the scope of that term or terms. Language

designating such nonlimiting examples and illustrations includes, but is not limited to: “for example,” “for instance,” “e.g.,” “in one embodiment.”

(8) Embodiments of the present invention can be implemented in a computer communicatively coupled to a network (for example, the Internet, an intranet, an internet, a WAN, a LAN, a SAN, etc.), another computer, or in a standalone computer. As is known to those skilled in the art, the computer can include a central processing unit (CPU) or processor, at least one read-only memory (ROM), at least one random access memory (RAM), at least one HD, and one or more input/output (I/O) device(s). The I/O devices can include a keyboard, monitor, printer, electronic pointing device (for example, mouse, trackball, stylus, etc.), or the like. In embodiments of the invention, the computer has access to at least one database over the network.

(9) ROM, RAM, and HD are computer memories for storing computer-executable instructions executable by the CPU or capable of being compiled or interpreted to be executable by the CPU. Within this disclosure, the term “computer readable medium” is not limited to ROM, RAM, and HD and can include any type of data storage medium that can be read by a processor. For example, a computer-readable medium may refer to a data cartridge, a data backup magnetic tape, a floppy diskette, a flash memory drive, an optical data storage drive, a CD-ROM, ROM, RAM, HD, or the like. The processes described herein may be implemented in suitable computer-executable instructions that may reside on a computer readable medium (for example, a disk, CD-ROM, a memory, etc.). Alternatively, the computer-executable instructions may be stored as software code components on a DASD array, magnetic tape, floppy diskette, optical storage device, or other appropriate computer-readable medium or storage device.

(10) In one exemplary embodiment of the invention, the computer-executable instructions may be lines of C++, Java, JavaScript, HTML, or any other programming or scripting code. Other software/hardware/network architectures may be used. For example, the functions of the present invention may be implemented on one computer or shared among two or more computers. In one embodiment, the functions of the present invention may be distributed in the network.

Communications between computers implementing embodiments of the invention can be accomplished using any electronic, optical, radio frequency signals, or other suitable methods and tools of communication in compliance with known network protocols.

(11) Additionally, the functions of the disclosed embodiments may be implemented on one computer or shared/distributed among two or more computers in or across a network.

Communications between computers implementing embodiments can be accomplished using any electronic, optical, radio frequency signals, or other suitable methods and tools of communication in compliance with known network protocols. It will be understood for purposes of this disclosure that a service is one or more computer processes, computing devices or both, configured to perform one or more functions. A service may present one or more interfaces which can be utilized to access these functions. Such interfaces include APIs, interfaces presented for a web services, remote procedure calls, remote method invocation, etc.

(12) Before discussing specific embodiments, a brief overview of the context of the disclosure may be helpful. As discussed above, it is desirable to give these users the ability to continue sending and receiving faxes without having to maintain local access to the publicly switched telephone network (PSTN).

(13) To that end, attention is now directed toward the embodiments of systems and methods for the sending, delivery and receiving of faxes disclosed herein. In particular, certain embodiments include a fax connector that may be deployed at a user's site, and a faxing system that may be deployed remotely from the user's site. The fax connector can be accessed at the user's site over a computer based network in order to perform functions associated with faxing, including sending, receiving and obtaining status on faxes. The fax connector deployed at the user's site communicates with the remotely deployed faxing system to send, receive, obtain status on, or perform other functions associated with, faxing.

(14) Because, in most cases, fax users already perform faxing utilizing fax servers, fax appliances, fax software, etc. connected to an computer based network (e.g. an IP (or other protocol) based network), faxing can be accomplished using embodiments of such systems and methods substantially without deviation from the user's previous faxing methodologies or workflows. Thus, such users can use their familiar fax environment without the restrictions of using a browser and without the need to maintain local telephony hardware, licenses, or telephone lines.

(15) Turning now to FIG. 1, one embodiment of an architecture of just such a faxing system is depicted. Fax architecture **100** includes a fax connector **106**, fax intermediary system **130** and fax sender/receiver (s/r) system **140** communicating over a computer based network **110** which may be, for example, the Internet, an internet, an intranet, a LAN, a WAN, an IP based network, etc. Fax intermediary system **130** has access to a data store **136**. Fax s/r system **140** includes one or interface to PSTN **120** and is configured to send or receive faxes over the PSTN **120**.

(16) Fax connectors **106** are installed at a site where users wish to send faxes. Users **102** in such an architecture **100** may be companies, individual users, or any other entity wishing to send or receive faxes. Accordingly, fax connector **106** may be one or more computing devices or one or more computer processes executing on one or more computer devices utilized by such users **102** to send or receive faxes. Fax connector **106** provides an interface such as an API or hardware interface through which a fax to send may be received from a client **104** of the fax connector **106**. These clients **104** may include a fax machine, a printer, an enterprise resource planning (ERP) system, etc. The fax connector **106** may communicate with fax intermediary system **130** over network **110** to send such a receive fax or to receive faxes.

(17) Fax intermediary system **130** may be one or more computing devices comprising a fax connector facing service **132** and a fax s/r facing service **134**. The fax connector facing service **132** may be configured to receive a fax; store the fax in data store **136**; report on a status of a previously received fax to be sent; determine if there are any faxes associated with a requestor in the data store **136**; obtain faxes from the data store **136**; and return one or more of these faxes to a requestor. Thus, fax connector facing service **132** may present an interface through which a request to send a fax may be received along with the document and phone number associated with that fax; an interface through which a request to check the status of a fax may be received and a status returned to the requestor; and an interface by which a request to determine if there are any faxes associated with a requestor may be received and one or more faxes associated with that requestor returned.

(18) Fax s/r facing service **134** may be configured to determine a fax to send from data store **136**; obtain and return a fax from the data store **136**; receive a status on a sent fax; update the status of a sent fax in data store **136**; receive a fax; and store a received fax in the data store **136**. Accordingly, fax service **134** may present an interface through which a request to determine if there are any faxes to send may be received and one or more faxes returned; an interface through which a request to update the status on a fax may be received along with data identifying the fax and the status on the fax; and an interface through which a received fax can be received including the document and one or more phone numbers associated with the received fax.

(19) Fax s/r facing service **134** may communicate with fax s/r system **140** over network **110** to send or receive a fax. Fax s/r system **140** includes fax s/r service **142**. Fax s/r service **142** is configured to request a fax to send at a certain time interval and send any faxes received in response to this request over the PSTN **120**; update the status of a sent fax; receive a fax over the PSTN **120**; and upload a received fax.

(20) Accordingly, when a user **102** in architecture **100** wishes to send a fax, he may utilize a client **104** at the user's site to send the fax. Client **104** provides the fax to the fax connector **106** through the interface provided by the fax connector **106**. Fax connector **106** then sends a request to fax connector facing service **132** to send the fax. Fax connector facing service **132** stores the fax received from the fax connector **106** in the data store **136**. At some point, fax s/r service **142** sends a request to fax s/r facing service **134** to determine if there are any faxes to send. Fax s/r facing

service **134** may receive the request, determine that there is a fax stored in the data store **136** to be sent and return this fax to the fax s/r service **142**. Fax s/r service **142** may receive this fax and send the fax to its destination over the PSTN **120**.

(21) During the sending of the fax, or thereafter, the fax s/r service **142** may send a status of the fax to fax s/r facing service **134**. Fax s/r facing service **134** may then update the status of the fax in the data store **136**. At some point, then, the fax connector **106** may send a request to the fax connector facing service **132** inquiring as to the status of the fax. Fax connector facing service **132** may obtain the status of the fax from the data store **136** and return it to the fax connector **106**, which may present it to a user at the user's site.

(22) Similarly, a fax may be received at fax s/r service **142** over PSTN **120**. When such a fax is received, fax s/r service **142** may send the fax to the fax s/r facing service **134**. Fax s/r facing service **134** may determine the fax connector **106** associated with the fax and store the fax in the data store **136**. At some point, then, the associated fax connector **106** may send a request to the fax connector facing service **132** to determine if there are any faxes associated with the fax connector **106**. Fax connector facing service **132** may obtain the associated fax from the data store **136** and return the fax to the fax connector **106**, which may then present the received fax to a user through the interface provided by the fax connector **106**.

(23) While it should be understood from a review of the above description that embodiments as disclosed may be implemented using a wide variety of architectures and hardware or software systems; that a fax intermediary system and a fax s/r system may be one or multiple systems that may or may not be distributed; that all of the services presented may or may not be one or multiple services with one or more interfaces; and that these one or more services may be implemented and accessed in a variety of manners, it will be useful here to describe a particular embodiment in more detail.

(24) As discussed above, in some cases it is desirable to remove telephony from a user's site. However, embodiments of the systems and methods disclosed herein may have many users. Thus, entities implementing or administering embodiments of the systems and methods presented herein may be responsible for multiple phone lines associated with multiple users. As phone lines require a relatively large amount of infrastructure, it may be desirable to centralize management of the telephone lines associated with these users. As the users themselves may be geographically distributed, it is also desirable to maximize such user's access to the fax intermediary system to reduce downtime and allow substantially simultaneous access from multiple users to the fax intermediary system.

(25) Therefore, in one embodiment, an entity implementing embodiments of the systems and method disclosed herein may implement the fax s/r system at a location over which they maintain control while the fax intermediary system may be implemented in a cloud based computing environment. By maintaining control of the access to the PSTN at a central location under the control of the entity, user's security concerns about access to the PSTN may be assuaged.

(26) The service or services of the fax intermediary system may, however, be deployed using a cloud based service provider, such as Microsoft Windows Azure, Amazon EC2, VMware vCloud, Google's cloud, etc. By deploying the services of the fax intermediary system in the cloud, such services can utilize the storage and functionality provided by the cloud provider while achieving increased accessibility and reliability and simultaneously alleviating the entity providing embodiments of such faxing systems and methods from the need to maintain the hardware or connectivity necessary for the provisioning of such services.

(27) With that in mind, turning now to FIG. 2, one embodiment of an architecture of a faxing system where certain services are deployed in a cloud based computing environment is depicted. Fax architecture **200** includes a network **270** over which a fax connector **206**, fax s/r system **240** and a set of fax services deployed in a cloud based computing environment may communicate.

(28) Fax connector **206** is provided at a site where users wish to send and receive faxes and may

have an associated identifier (i.e. the fax connector identifier). One or more telephone numbers associated with a user's fax line are configured such that any calls to those numbers are received over PSTN **220** at phone lines coupled to fax s/r system **240**. Thus, providers of embodiments as depicted herein may notify or otherwise arrange with telephony providers, to configure one or more telephone numbers associated with the user employing fax connector **206** such that calls to these numbers received at telephone lines of PSTN coupled to fax s/r system **240**.

(29) Fax connector **206** provides an interface **202** through which a client of the fax connector **206** may send faxes, receive faxes, check the status on sent faxes, etc. Fax connector **206** comprises three services: send fax service **204**, status service **205** and receive service **208**. Send fax service **204** may be configured to receive a document and a phone number from the interface, convert the document to a particular format (e.g. Tagged Image File Format (TIFF)) if desired, send a request that the fax be sent and receive an identifier for that fax in return. Status service **205** may be configured to send a request for status on a sent fax along with a fax identifier, receive the status of the fax in return and present this status to a user through interface **202**. Receive service **208** may be configured to send a request to identify any faxes received for the user along with the fax connector identifier for the fax identifier **206** and receive one or more faxes returned in response to this request.

(30) Cloud **244** may be a cloud based computing platform such as Microsoft Windows Azure Platform or the like. The set of faxing services deployed in the cloud include a connector facing send **232**, connector facing receive **234**, a fax service facing receive **236** and a fax service facing send **238**. These services may have one or more interfaces that may be accessed over network **270** using hypertext transfer protocol (http) calls or other types of network protocols or interfaces.

(31) The cloud **244** may also provide certain services that may be utilized by the deployed set of faxing services, such services may include, for example the Microsoft Windows Azure Storage services such as table storage **252**, queue storage **254** or other forms of storage not depicted such as blob storage, etc. This storage may be used during operation of the set of service and may store various configuration information associated with the services. For example, queue storage **254** may be used to store entries associated with faxes to send, where each entry in the queue is associated with a job identifier for that fax, while table storage **252** may be used to store a mapping table, a send fax table and a received fax table.

(32) The mapping table may associate phone numbers with fax connector identifiers. Thus, for example, if a user's telephone number has been configured such that calls to this number is received at telephone lines of PSTN **220** coupled to fax s/r system **240**, there may be an entry in the mapping table associating this phone number with the fax connector identifier for that user's fax connector **206**. The send fax table may associate job identifiers associated with entries in the queue with various data associated with a corresponding fax, including the document of the fax, the telephone number where the fax is to be sent, a fax connector identifier associated with the fax, a status associated with the fax, or other information. The received fax table may also be stored in the table storage **252**, where the received fax table may include the document of the received fax, a fax connector identifier associated with fax connector where the fax is to be sent, the telephone number from which the fax was received, or other information. Other services provided by the cloud **244** (not depicted) may also be utilized (e.g. in the case of Microsoft Windows Azure Platform: Web Role, Worker Role, VM Role, Access Control, Content Delivery Network, Caching, etc. Other services in other cloud based computing platforms may also be utilized).

(33) The connector facing send **232** may have an interface for sending faxes through which a request to send a fax may be received. Such a request may include the document to be faxed (which may be in TIFF format or another electronic format) and a phone number where the fax is to be sent. When such a request is received, the connector facing send **232** may enqueue that fax in queue storage **254**, store data associated with the fax in the table storage **252** and return an identifier of the fax (i.e. a fax identifier) to the fax connector **206**. Connector facing send **232** may

additionally have an interface through which a status inquiry may be received along with a fax identifier. In response, connector facing send **232** may access the table storage **252** to read a status of the fax corresponding to the fax identifier from the table storage **252** and return the status of the fax to the requestor.

(34) Connector facing receive **234** may have an interface through which a request for received faxes may be received along with a fax connector identifier. In response, the connector facing receive **234** may access table storage **252** to determine if there are any received faxes associated with the fax connector identifier. One or more of these faxes may be returned to the requestor by connector facing receive **234**.

(35) Fax service facing send **238** may have an interface through which a request for faxes to send may be received. In response, the fax service facing send **238** may access queue storage **254** to determine if there are any faxes waiting to be sent and select one or more of these faxes based on a priority. As will be noted, queue storage **254** may be a type of queue (e.g. Microsoft Messaging Queue (MSMQ), IBM MQSeries, etc.) that allows priority to be implemented in conjunction with the queue. Thus, entries in the queue associated with faxes may be selected based on almost any priority desired, for example, timing, quality of service, source or destination, or some other criteria. When a fax has been selected, the fax identifier may be used to access the fax in the table storage **252** and the fax returned to the requestor.

(36) The fax service facing send **238** may also have an interface through which a request to update the status of the fax may be received along with a status and a fax identifier. The fax service facing send **238** may then update the status of the fax in the table store **252**.

(37) Fax service facing receive **236** may have an interface through which a request to upload a received fax may be received, along with the document and telephone number from which the fax was received. When such a request is received, the fax service facing receive **236** may access the table storage **252** to determine a fax connector identifier associated with the phone number from which the fax was received and store the fax in association with the fax connector identifier in the table storage **252**.

(38) Fax service **242** is configured to request a fax to send at a certain time intervals, receive a fax identifier, a document, and an associated telephone number in response and send the received fax to the telephone number over the PSTN **220**. During the sending of the fax, the fax service **242** can send a request to update the status of the fax along with the fax identifier and an updated status. Such a request may be sent one or more times while the fax service is sending the fax over the PSTN **220** (e.g. a request may be sent after each page is sent, after the entire fax is completed, etc.). Fax service **242** may also receive faxes over PSTN **220**. When a fax is received, the fax service **242** may send a request to upload the fax along with the document and the telephone number from which that fax was received.

(39) As can be seen then, the fax connector, fax s/r system and the set of fax services deployed in the cloud may interact to send, deliver and receive faxes. Embodiments of this process may be illustrated in more detail with reference to FIGS. **3A1-3B**. Referring first to FIG. **3A1-3A3**, one embodiment of a method for the sending of a fax employing an architecture including cloud based fax services is depicted. Initially at step **302**, a user wishing to send a fax may provide a document and an associated phone number as a destination for the document. The document may be received by the send service of the fax connector at the user's site and, at step **304**, may be converted (e.g. to a TIFF or another format) by the send fax service of the fax connector at the user's site.

Alternatively, the document may not be converted, or may be converted at another location. The send fax service of the user's fax connector can then send a request to the connector facing send service deployed in the cloud at step **306** requesting that a fax be sent. The request may include the document and the destination phone number (or these may be sent subsequently).

(40) When this send request is received by the connector facing send service in the cloud at step **308**, the connector facing send service may enqueue a job in queue storage for the sending of the

fax associated with the received send request at step **310**. When the job is enqueued, the queue storage in the cloud may return a job identifier associated with the created entry in queue storage at step **312**. This job identifier is utilized by the connector facing send service to create an entry in table storage at step **314**, where the entry associates the job identifier for the fax with the document and phone number provided in the request. The connector facing send service can then reply to the send service of the fax connector with the job identifier at step **316**. The send service of fax connector may receive and store this job identifier at step **318** such that it may be utilized in subsequent processing, such as for example, a subsequent status inquiry.

(41) At some point then, at step **319**, the fax service may send a request to the fax service facing send service to determine if there are any faxes to be sent. Fax service facing send service may receive this request at step **320**, access the queue storage to determine if there are any job entries in the queue storage and determine one or more of these jobs at step **322**. As discussed above, this determination may be based on a priority associated with entries in the queue storage.

(42) If there is a job to be sent, the job identifier of the determined entry may be determined from the entry in the queue. This job identifier may be utilized to access the entry in the table storage associated with that job identifier and the document and phone number associated with the job identifier read from the table at step **324** and returned to the fax service at step **326**. In some cases, the job identifier may also be sent to the fax service to be utilized for subsequent processing, for example, status updates and the like.

(43) The fax service may receive the fax at step **328** and send the fax to the phone number over the PSTN at step **330**. During the sending of the fax over the PSTN, the fax service may, at step **332**, send one or more requests to the fax service facing send service to update the status of the fax during or after sending of the fax over the PSTN. This request may include a job identifier of the fax and a status (such as transmitted, failed, o.k., etc.) or an associated time. The fax service can also update a reporting service at step **334**, such that data on sent faxes may be accessed or otherwise processed.

(44) Fax service facing send service may receive the status update request at step **336** and update the entry in the table storage associated with the job identifier with the received statuses at step **338**.

(45) At some point then, status service of the user's fax connector may send a status query request to the connector facing send service at step **340**. This request may contain a previously provided job identifier as discussed above. The connector facing send service may receive the status inquiry request at step **342**, access the table storage to determine the job status at step **344** and return the job status at step **346**. The status service of the fax connector can then receive and display the job status to the user at step **348**.

(46) Moving now to FIG. 3B, one embodiment of a method for the reception of a fax employing an architecture including cloud based fax services is depicted. As discussed above, the PSTN is configured such that calls on one or more phone numbers associated with a particular fax connector are received at the fax service. Similarly, then, a table may exist in the table storage of the cloud based fax service that has entries mapping these phone numbers to the fax connector identifier associated with that fax connector.

(47) Initially, then, at step **352**, the fax service may receive a fax associated from a phone number over the PSTN. The fax service may then send a request to upload a received fax to the fax service facing receive service at step **354**. This request may include the document received along with the phone number from which it was received and the phone number of the fax's intended destination (or the document and phone number may be sent at a subsequent time). The fax service can also update a reporting service at step **356**, such that data on received faxes may be accessed or otherwise processed.

(48) The fax service facing receive may receive the request to upload the fax at step **358**. At step **360**, the fax service facing receive service may then access the table mapping phone numbers to fax connector identifiers to determine the fax connector identifier associated with the destination phone

number. The fax is then stored in an entry in table storage in association with the associated fax connector identifier at step **362**. Other information may also be associated with the entry, including, for example, the time received, the phone number from which it was sent, the destination phone number, etc. If an associated fax connector identifier cannot be found, the fax may be stored in a dead letter table and the reporting data updated.

(49) At some point then, at step **364**, the receive service of the fax connector may send a request to the connector facing receive service. This request may identify the fax connector using the fax connector identifier for that fax connector. When this request is received at the connector facing receive service at step **366**, the connector facing receive service determines at step **368** if there are any faxes associated with that fax connector identifier of the request by accessing the table storage to determine if there are any entries in the received fax table associated with that fax connector identifier. If any entries are found in the table storage one or more of the faxes may be returned to the receive service at step **370**. The determination of which of the faxes to return may be based on almost any criteria desired, such as time received or the like. Additionally, multiple faxes may be returned in response to a received request. More specifically, the connector facing receive service may access an entry associated with the fax connector identifier to obtain the document and telephone number to which the fax was sent or from which the fax was received. This document along with one or more of the phone numbers may be returned to the receive service of the fax connector.

(50) When the receive service of the fax connector receives the fax at step **372**, it may display the received fax at step **374** through its interface. This may entail displaying the fax through a computer display, printing the fax on a client device or some other method of display entirely.

(51) As can be seen then, using embodiments of the systems and methods disclosed herein, a user can use their familiar fax environment without the restrictions of using a browser and without the need to maintain local hardware, licenses, or telephone lines. However, the user can see status updates associated with outbound faxes in their local system as they would using conventional techniques. Additionally, received faxes may come into the user's local system leveraging the rules and workflows to which they are accustomed.

(52) Although the invention has been described with respect to specific embodiments thereof, these embodiments are merely illustrative, and not restrictive of the invention. The description herein of illustrated embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise forms disclosed herein (and in particular, the inclusion of any particular embodiment, feature or function is not intended to limit the scope of the invention to such embodiment, feature or function). Rather, the description is intended to describe illustrative embodiments, features and functions in order to provide a person of ordinary skill in the art context to understand the invention without limiting the invention to any particularly described embodiment, feature or function. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes only, various equivalent modifications are possible within the spirit and scope of the invention, as those skilled in the relevant art will recognize and appreciate. As indicated, these modifications may be made to the invention in light of the foregoing description of illustrated embodiments of the invention and are to be included within the spirit and scope of the invention. Thus, while the invention has been described herein with reference to particular embodiments thereof, a latitude of modification, various changes and substitutions are intended in the foregoing disclosures, and it will be appreciated that in some instances some features of embodiments of the invention will be employed without a corresponding use of other features without departing from the scope and spirit of the invention as set forth. Therefore, many modifications may be made to adapt a particular situation or material to the essential scope and spirit of the invention.

(53) Reference throughout this specification to “one embodiment,” “an embodiment,” or “a specific embodiment” or similar terminology means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment and may not

necessarily be present in all embodiments. Thus, respective appearances of the phrases “in one embodiment,” “in an embodiment,” or “in a specific embodiment” or similar terminology in various places throughout this specification are not necessarily referring to the same embodiment. Furthermore, the particular features, structures, or characteristics of any particular embodiment may be combined in any suitable manner with one or more other embodiments. It is to be understood that other variations and modifications of the embodiments described and illustrated herein are possible in light of the teachings herein and are to be considered as part of the spirit and scope of the invention.

(54) In the description herein, numerous specific details are provided, such as examples of components and/or methods, to provide a thorough understanding of embodiments of the invention. One skilled in the relevant art will recognize, however, that an embodiment may be able to be practiced without one or more of the specific details, or with other apparatus, systems, assemblies, methods, components, materials, parts, and/or the like. In other instances, well-known structures, components, systems, materials, or operations are not specifically shown or described in detail to avoid obscuring aspects of embodiments of the invention. While the invention may be illustrated by using a particular embodiment, this is not and does not limit the invention to any particular embodiment and a person of ordinary skill in the art will recognize that additional embodiments are readily understandable and are a part of this invention.

(55) Any suitable programming language can be used to implement the routines, methods or programs of embodiments of the invention described herein, including C, C++, Java, assembly language, etc. Different programming techniques can be employed such as procedural or object oriented. Any particular routine can execute on a single computer processing device or multiple computer processing devices, a single computer processor or multiple computer processors. Data may be stored in a single storage medium or distributed through multiple storage mediums, and may reside in a single database or multiple databases (or other data storage techniques). Although the steps, operations, or computations may be presented in a specific order, this order may be changed in different embodiments. In some embodiments, to the extent multiple steps are shown as sequential in this specification, some combination of such steps in alternative embodiments may be performed at the same time. The sequence of operations described herein can be interrupted, suspended, or otherwise controlled by another process, such as an operating system, kernel, etc. The routines can operate in an operating system environment or as stand-alone routines. Functions, routines, methods, steps and operations described herein can be performed in hardware, software, firmware or any combination thereof.

(56) Embodiments described herein can be implemented in the form of control logic in software or hardware or a combination of both. The control logic may be stored in an information storage medium, such as a computer-readable medium, as a plurality of instructions adapted to direct an information processing device to perform a set of steps disclosed in the various embodiments. Based on the disclosure and teachings provided herein, a person of ordinary skill in the art will appreciate other ways and/or methods to implement the invention.

(57) It is also within the spirit and scope of the invention to implement in software programming or of the steps, operations, methods, routines or portions thereof described herein, where such software programming or code can be stored in a computer-readable medium and can be operated on by a processor to permit a computer to perform any of the steps, operations, methods, routines or portions thereof described herein. The invention may be implemented by using software programming or code in one or more general purpose digital computers, by using application specific integrated circuits, programmable logic devices, field programmable gate arrays, optical, chemical, biological, quantum or nanoengineered systems, components and mechanisms may be used. In general, the functions of the invention can be achieved by any means as is known in the art. For example, distributed, or networked systems, components and circuits can be used. In another example, communication or transfer (or otherwise moving from one place to another) of

data may be wired, wireless, or by any other means.

(58) A “computer-readable medium” may be any medium that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, system or device. The computer readable medium can be, by way of example, only but not by limitation, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, system, device, propagation medium, or computer memory. Such computer-readable medium shall generally be machine readable and include software programming or code that can be human readable (e.g., source code) or machine readable (e.g., object code).

(59) A “processor” includes any, hardware system, mechanism or component that processes data, signals or other information. A processor can include a system with a general-purpose central processing unit, multiple processing units, dedicated circuitry for achieving functionality, or other systems. Processing need not be limited to a geographic location, or have temporal limitations. For example, a processor can perform its functions in “real-time,” “offline,” in a “batch mode,” etc. Portions of processing can be performed at different times and at different locations, by different (or the same) processing systems.

(60) It will also be appreciated that one or more of the elements depicted in the drawings/figures can also be implemented in a more separated or integrated manner, or even removed or rendered as inoperable in certain cases, as is useful in accordance with a particular application. Additionally, any signal arrows in the drawings/figures should be considered only as exemplary, and not limiting, unless otherwise specifically noted.

(61) Furthermore, the term “or” as used herein is generally intended to mean “and/or” unless otherwise indicated. As used herein, a term preceded by “a” or “an” (and “the” when antecedent basis is “a” or “an”) includes both singular and plural of such term (i.e., that the reference “a” or “an” clearly indicates only the singular or only the plural). Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

(62) Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any component(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature or component.

Claims

1. A method for faxing, comprising: at a first service in a faxing platform: receiving a first request for sending a first fax, the first fax associated with a first document; storing the first fax at a data store on the faxing platform; at a second service in the faxing platform: determining the first fax is in the data store of the faxing platform; sending the first fax to an associated destination over a publicly switched telephone network (PSTN) or a computer based network based on the associated destination.
2. The method of claim 1, wherein the first request for sending the first fax was received from a fax connector at a site of a user, the fax connector associated with the faxing platform.
3. The method of claim 1, wherein sending the first fax to the associated destination comprises requesting, from the second service, a third service to send the first fax.
4. The method of claim 1, further comprising: at the second service: receiving a status associated with the first fax; and updating the first fax in the data store with the status.
5. The method of claim 4, wherein the status is received from the third service.
6. The method of claim 3, further comprising: at the second service: receiving a second fax in association with the faxing platform; associating the second fax with the fax connector at the site of the user; storing the second fax in the data store on the faxing platform; at the first service: receiving a second request from the fax connector at the site of the user, the second request requesting any faxes associated with the fax connector; determining the second fax in the data store

is associated with the fax connector; and providing the second fax to the fax connector in response to the second request.

7. The method of claim 6, wherein the second fax is received at the third service.

8. A non-transitory computer readable medium storing instructions for: at a first service in a faxing platform: receiving a first request for sending a first fax, the first fax associated with a first document; storing the first fax at a data store on the faxing platform; at a second service in the faxing platform: determining the first fax is in the data store of the faxing platform; sending the first fax to an associated destination over a publicly switched telephone network (PSTN) or a computer based network based on the associated destination.

9. The non-transitory computer readable medium of claim 8, wherein the first request for sending the first fax was received from a fax connector at a site of a user, the fax connector associated with the faxing platform.

10. The non-transitory computer readable medium of claim 8, wherein sending the first fax to the associated destination comprises requesting, from the second service, a third service to send the first fax.

11. The non-transitory computer readable medium of claim 8, further comprising instructions for: at the second service: receiving a status associated with the first fax; and updating the first fax in the data store with the status.

12. The non-transitory computer readable medium of claim 11, wherein the status is received from the third service.

13. The non-transitory computer readable medium of claim 10, further comprising instructions for: at the second service: receiving a second fax in association with the faxing platform; associating the second fax with the fax connector at the site of the user; storing the second fax in the data store on the faxing platform; at the first service: receiving a second request from the fax connector at the site of the user, the second request requesting any faxes associated with the fax connector; determining the second fax in the data store is associated with the fax connector; and providing the second fax to the fax connector in response to the second request.

14. The non-transitory computer readable medium of claim 13, wherein the second fax is received at the third service.

15. A system, comprising: a processor; and a non-transitory computer readable storage medium storing instructions for: at a first service in a faxing platform: receiving a first request for sending a first fax, the first fax associated with a first document; storing the first fax at a data store on the faxing platform; at a second service in the faxing platform: determining the first fax is in the data store of the faxing platform; sending the first fax to an associated destination over a publicly switched telephone network (PSTN) or a computer based network based on the associated destination.

16. The system of claim 15, wherein the first request for sending the first fax was received from a fax connector at a site of a user, the fax connector associated with the faxing platform.

17. The system of claim 15, wherein sending the first fax to the associated destination comprises requesting, from the second service, a third service to send the first fax.

18. The system of claim 15, wherein the instructions further comprise instructions for: at the second service: receiving a status associated with the first fax; and updating the first fax in the data store with the status.

19. The system of claim 18, wherein the status is received from the third service.

20. The system of claim 17, wherein the instructions further comprise instructions for: at the second service: receiving a second fax in association with the faxing platform; associating the second fax with the fax connector at the site of the user; storing the second fax in the data store on the faxing platform; at the first service: receiving a second request from the fax connector at the site of the user, the second request requesting any faxes associated with the fax connector; determining the second fax in the data store is associated with the fax connector; and providing the

second fax to the fax connector in response to the second request.

21. The system of claim 20, wherein the second fax is received at the third service.
